



Management Science

Publication details, including instructions for authors and subscription information:
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To cite this article:

Hayri A. Arslan, Necati Tereyağoglu, Övünç Yılmaz (2023) Scoring a Touchdown with Variable Pricing: Evidence from a Quasi-Experiment in the NFL Ticket Markets. *Management Science* 69(8):4435-4456. <https://doi.org/10.1287/mnsc.2022.4588>

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Scoring a Touchdown with Variable Pricing: Evidence from a Quasi-Experiment in the NFL Ticket Markets

Hayri A. Arslan,^{a,*} Necati Tereyağolu,^{b,*} Övünç Yılmaz^{c,*}

^a Alvarez College of Business, University of Texas at San Antonio, San Antonio, Texas 78249; ^b Darla Moore School of Business, University of South Carolina, Columbia, South Carolina 29208; ^c Leeds School of Business, University of Colorado, Boulder, Colorado 80309

*Corresponding author

Contact: hayri.arslan@utsa.edu,  <https://orcid.org/0000-0002-2359-7773> (HAA); necati.tereyagolu@moore.sc.edu,  <https://orcid.org/0000-0002-2824-0036> (NT); ovunc.yilmaz@colorado.edu,  <https://orcid.org/0000-0002-0746-353X> (ÖY)

Received: September 3, 2019

Revised: October 4, 2020; August 9, 2021;
December 21, 2021

Accepted: March 18, 2022

Published Online in Articles in Advance:
October 22, 2022

<https://doi.org/10.1287/mnsc.2022.4588>

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Abstract. Although variable pricing enables sports teams to respond to changing demand across games, reports point to somewhat limited implementation. In this paper, we study the implications of a switch to variable pricing on the primary and resale ticket markets using quasi-experimental data from the National Football League. Applying a difference-in-differences technique with propensity score weighting, we first show that the adoption of variable pricing increases primary market ticket sales by 1.59% per game. We then explore the mechanism behind this positive impact. We find that lower prices for the less attractive games are well received by customers, and customers do not have an unfavorable response to higher prices for the more attractive games. Our analysis of the customers' activities in the resale market shows that the minimum resale market prices increase for the more attractive games following the switch to variable pricing. This analysis provides evidence for the increase in the option value created by the resale market and explains the positive response to higher prices for the more attractive games. Finally, we investigate when variable pricing is likely to be the most effective. We find that variable pricing has greater success in team hometowns with lower income levels and higher income diversity, and creating more game tiers in variable pricing leads to higher sales.

History: Accepted by Gabriel Weintraub, revenue management and market analytics.

Supplemental Material: The data files and online appendix are available at <https://doi.org/10.1287/mnsc.2022.4588>.

Keywords: ticket pricing • resale markets • sports events • difference-in-differences • propensity score

1. Introduction

Gate revenues, the primary market ticket sales for live sporting events, are a major revenue source for sports teams. In the North American sports market alone, the gate revenues generated in 2018 were estimated as \$19.2 billion (PwC 2019). Sports teams often have the power to price their home game tickets that are sold in different forms (e.g., season tickets, single-game tickets) and aim at developing a pricing strategy that leverages the full potential of this revenue source. Because customers value each seat (based on view angle, distance to the field, etc.) and each game (based on opponent team, time of the year, etc.) differently, teams use pricing strategies such as *seat-based pricing*, where different seat categories are offered at different prices, and *variable pricing*, where ticket prices vary across games.

Although seat-based pricing has been very popular in the entertainment industry (Courty 2015) and its benefits are well documented (Leslie 2004, Courty and Pagliero 2012), reports point to somewhat limited implementation of variable pricing (Engber 2017). On

one hand, variable pricing enables teams to respond to changing demand across games compared with a fixed price strategy that favors meeting the demand for the most popular games and leaves others underused (Phillips 2005). On the other hand, variable pricing for sports games is a complex revenue management problem that involves price decisions that are prone to uncertainty. In particular, the unpredictable nature of the popularity of a sports game (Xu et al. 2019) and fluctuating valuation of customers' experience over the course of the season (Deloitte 2013) may cause sports teams to hesitate. In addition, sports tickets can be resold on the resale market, and how customers react to price changes through variable pricing is unclear: The lower primary market prices for games that are deemed less attractive may lead to increased cannibalization of primary market ticket sales (Talluri and van Ryzin 2004) if the resale market prices decrease even more compared with previous years. Conversely, higher primary market prices for games that are deemed more attractive by sports teams may actually lead to decreased demand for these

games because of diminished arbitrage opportunities (Leslie and Sorensen 2014). Thus, it is important for the sports teams to understand the implications of variable pricing. In this paper, we empirically examine the impact of variable pricing in a quasi-experimental setting and seek to answer a fundamental question: How are sports teams' primary market ticket sales affected when they switch from fixed to variable pricing?

The conventional wisdom would predict that a decrease (an increase) in the face value of a ticket for a less (more) attractive game through variable pricing leads to higher (lower) primary market ticket sales. However, in the presence of a resale market, an answer to this question requires an understanding of customers' actions in resale markets in addition to their actions in the primary market. The resale market acts as a substitute for the primary market and helps customers in different ways while creating an interesting trade-off for the sports teams: First, customers may recover a substantial portion of their investment by reselling the ticket if they change their mind about going to a game or may actually make money if the game sells out and someone else is willing to pay more than the face value of the ticket. This value created by the resale market is called the *option value* and increases the sport teams' ability to charge higher prices (Bennett et al. 2015). Second, a customer may find tickets well below face value on the resale market if the sellout is unlikely or may find a ticket for a game in the last-minute even if the game sells out. However, this availability may lead customers to defer their ticket purchases to the resale market, resulting in *cannibalization* of primary market ticket sales (Bennett et al. 2015). Due to this interplay between two markets, customers' decisions in the primary market depend on their expectation about the resale market (Leslie and Sorensen 2014), and it is not clear a priori how variable pricing would affect the primary market ticket sales. Thus, we seek to answer another important question that helps us to understand the mechanism behind the impact of variable pricing on the primary market ticket sales: How do price changes through variable pricing affect customer decision-making in primary and resale markets?

To answer these questions, we assemble a novel data set that includes the game level primary market ticket sales, as well as the minimum ticket prices and number of ticket listings on the resale market before and after the period when the National Football League (NFL) in the United States first allowed teams to use variable pricing strategies, that is, just before the 2014 season. After the announcement, 15 of the 32 NFL teams quickly switched from fixed to variable pricing. The following season, three more teams joined these teams. The main challenge in measuring the effects of this switch is that it would have been a strategic decision by these teams, thus potentially

not random. Hence, a simple comparison of the primary and resale market activities between home games of teams that switch to variable pricing and home games of those that do not may be misleading, especially if there are differences across teams or games (e.g., demographic differences across home game locations). To address this endogenous switch-selection threat to the identification of the causal effect of variable pricing, we use a propensity score weighted difference-in-differences strategy at the team-home game level. This approach ensures that treated and nontreated games are comparable in terms of pretreatment trends of primary and resale market activities and several important variables that might influence a team's decision to switch to variable pricing.

We contribute three substantial findings.

First, we document the positive impact of variable pricing on the primary market ticket sales in what constitutes the first causal analysis of this pricing strategy in the literature. We find that the adoption of variable pricing increases the primary market ticket sales by 1.59%, on average, per game. We further show that this positive effect flattens out after an initial jump following the switch to variable pricing. These results are encouraging for those sports teams that are hesitant about a switch to variable pricing.

Second, to provide a clearer picture, we explore the mechanism behind this positive impact. We find that the lower prices for less attractive games are well received by the customers. Particularly, the deeper the discount is, the more positive the impact on primary market ticket sales. On the other hand, customers do not have an unfavorable reaction to higher prices for more attractive games. Counterintuitively, the higher the price increase is, the more positive the impact on primary market ticket sales, likely because of an increase in the option value for these tickets. We check the validity of this explanation by presenting evidence from customers' activities in the resale market. The minimum ticket prices on the resale market do not change for the less attractive games and increase for the more attractive games. This observation confirms that (i) a decrease in the price of the ticket for less attractive games does not change the option value, therefore the lower primary market price drives the primary market ticket sales up, and (ii) an increase in the price of the ticket for more attractive games increases the option value and leads to higher sales in the primary market. We also find that the number of ticket listings on the resale market increases for less attractive games but does not change for more attractive games. Combining with the price changes in the resale market, these changes in the number of ticket listings suggest that variable pricing does not lead to a higher level of cannibalization of primary market ticket sales, nor does it pull customers that purchase tickets seeking arbitrage opportunities out of the market.

Third, we further investigate when variable pricing is likely to be the most effective. We find that variable pricing has greater success in team hometowns with lower income levels and higher income diversity. The discounts given for the less attractive games are particularly well received in these hometowns. In addition, using more game tiers in variable pricing leads to higher primary market ticket sales, and a complex seat-based price differentiation does not suffer from the adoption of variable pricing. These results reveal that it is important to invest in sports analytics personnel that could understand the market characteristics and formulate the right pricing strategy.

2. Literature Review

This paper is primarily related to two streams of literature: price discrimination and variable pricing, and secondary markets. Price discrimination across different seat categories is common in the entertainment industry. As the theory of second-degree price discrimination predicts (Pigou 1932), seat-based pricing essentially steers people with different preferences toward various seat categories and helps organizations maximize the surplus extracted from different customer bases. Rosen and Rosenfield (1997) show that the return from seat-based pricing depends on the relative variance of customer willingness-to-pay for each seat category. Courty and Pagliero (2012) complement this finding by providing empirical evidence from the North American concert industry. Using data from a Broadway theater, Leslie (2004) shows that a switch to seat-based pricing from uniform pricing can lead to revenue improvements without significant changes in aggregate customer welfare. We add to this stream of research by considering qualitative differences not just across seat categories but also across an organization's multiple events in a season.

Some event organizations recently switched to variable pricing for their multiple offerings across a season (Stevenson 2019). Similar pricing strategies have been implemented in a variety of settings: peak-load pricing by utility firms (Vickrey 1971, Kök et al. 2018), surge pricing on service platforms (Cohen et al. 2016, Cachon et al. 2017), congestion pricing for parking spaces (Feldman et al. 2022), and price discrimination with advance selling for entertainment offerings (Gale and Holmes 1993, Gallego and Şahin 2010, Cachon and Feldman 2017). However, the implementation of variable pricing is limited in the entertainment industry and its implications have been understudied. Arslan et al. (2022) develop a data-driven pricing tool for a college football team that uses variable pricing. Courty and Davey (2020) estimate the impact of variable pricing on Major League Baseball teams' annual revenues by exploiting a panel covering 1999–2017. We contribute to this line of research by (i) empirically documenting the positive

impact of variable pricing using quasi-experimental data from the NFL ticket markets, (ii) explaining the mechanism behind this positive impact, and (iii) offering insights on the determinants behind the success of variable pricing.

Our work is also related to the research on secondary markets. Because event tickets are often transferable, we need to consider how customers react to different pricing strategies in the presence of secondary markets. The well-established finding is that used-good markets cannibalize new product sales (Suslow 1986). However, active secondary markets can also benefit producers through indirect price discrimination opportunities (Anderson and Ginsburgh 1994, Hendel and Lizzeri 1999, Chen et al. 2013). In the context of ticket pricing, Bennett et al. (2015) characterize cannibalization and option value effects of resale markets on primary markets. The significant size of the resale market (Trefis Team 2017, Connolly and Krueger 2018), and the increasing availability of resale market data (Rishe 2014) also explain the recent academic interest in understanding customer behavior in the primary versus the resale market.

This research stream mainly explores the implications of a resale market. One side of this research stream focuses on showing how the presence of scalpers, who purchase tickets in advance from the primary market to resell at a higher price on the resale market, influence the event organizers' revenues. Su (2010) shows that organizations can increase their expected profits by selling tickets early and transferring the inventory risk to scalpers. Cui et al. (2014) identify conditions in which the presence of scalpers may have positive effects and suggest selling ticket options to increase revenue. On the same side of this research stream, but unlike the aforementioned papers, Sweeting (2012) focuses on modeling the pricing decisions of resellers in the resale market and shows that simple dynamic pricing models perform well in explaining their dramatic price cuts. Another side of this research stream aims to empirically show the effects of having a resale market on organizations' outcomes. Leslie and Sorensen (2014) show that a resale market can increase the allocative efficiency of seats by 5% in the rock concert industry. Lewis et al. (2019) show that a resale market increases season ticket sales for a Major League Baseball team. Zou and Jiang (2020) show that content producers, organizers, and customers can benefit from the existence of a resale market if primary ticket platforms control the resale market. Baliga and Ely (2013) and Bhave and Budish (2017) find that the use of auctions in ticket selling can alleviate the problem of arbitrage profits associated with underpriced tickets. All these findings show that a thorough understanding of the effects of a new pricing strategy requires a clear observation of both resale and primary market activities. Hence, in this paper, we do not just empirically show the effect of variable pricing on primary market ticket sales

but also explore customer decision making in the resale market.

3. Background, Empirical Context, and Data

In this section, we provide details about the NFL and its pricing strategy shift, and describe our data.

3.1. NFL and Ticket Markets

The NFL is a professional football league in the United States, consisting of 32 teams competing in two conferences: the National Football Conference (NFC) and the American Football Conference (AFC). Each conference has four divisions with four teams each. In a regular season, each team plays 16 games (8 home and 8 away) that run from early September to late December. A team's schedule is set using a formula: six games against three division rivals (one home game and one away game against each rival), eight games against all the opponents from one division from the NFC and one division from AFC on a rotating basis (one game for each), and two games within the conference based on their previous season performance. At the end of a season, six teams from each conference advance to the playoffs. In this study, we focus on the regular season games, because the pricing decisions for playoff games are conditional on a team's appearance and made right before the playoff games (as opposed to at the beginning of the season like regular season ticket pricing decisions).

NFL teams have the power to decide on the ticket pricing strategies for home games, and often sell two types of tickets: (i) season ticket packages, that is, ticket bundles that provide access to all games during a season; and (ii) single-game tickets, that is, tickets that provide access to a specific game. These tickets can be accessed through primary markets, such as the team website, box office, or authorized sellers such as Ticketmaster. Customers who purchase tickets from primary markets can sell their tickets through online resale ticket marketplaces such as StubHub.

3.2. Variable Pricing in the NFL

Across all major sports leagues in the United States, the NFL is known for holding on the longest to a fixed pricing strategy, that is, each home game is priced the

same. However, just prior to the 2014 season, the NFL allowed teams to individually decide whether to continue with fixed pricing or switch to variable pricing, in which they set varying prices across games. Major news outlets point to the superiority of variable pricing in setting the prices according to the expected popularity of games, in recouping value lost to the resale market, in providing more value and transparency to season ticket buyers, and, most importantly, in increasing sales in the primary market (Fisher 2014). These objectives also align with the league's goals to improve revenue by increasing the attendance, which has a direct impact on future television and sponsorship contracts (Phillips 2017), and by preventing potential television blackouts (Drayer et al. 2012).¹

Due to the short preparation time until the release of the coming season's game schedule in April 2014, only 15 of the 32 NFL teams were able to implement variable pricing for the 2014 season. The following season, three additional teams introduced variable pricing.² Table 1 provides examples of price points for some seat sections before and after the switch to variable pricing for the New England Patriots and the Buffalo Bills.

3.3. Data

We merge data from the following 10 sources to estimate the effect of the change from fixed to variable pricing on NFL ticket markets: (1) the NFL's website, (2) Internet Archive's Wayback Machine search tool, (3) Team Marketing Report, (4) Pro-Football-Reference.com, (5) SportsOddsHistory.com, (6) FiveThirtyEight.com, (7) TheSportsDaily.net, (8) Forbes.com, (9) the U.S. Census Bureau, and (10) Brandeis University Heller School for Social Policy and Management's DiversityData.org project.

Our primary data set contains information on the paid attendance of each regular season game in the NFL, which is from NFL.com. To obtain these data, we download all the NFL game books for each regular season game in 2012–2015.³ Brown (2011) and Bachman (2018) point out that it has become the norm for teams to report their actual ticket sales data as their paid attendance numbers. Hence, we use the paid attendance numbers in the NFL game books as our primary market ticket sales data.

Table 1. Examples of Price Points Under Fixed and Variable Pricing for Two NFL Teams

Team	Section	2013 Fixed price	2014		
			High price	Medium price	Low price
New England Patriots	Lower level	\$211	\$245	—	\$195
	Upper level	\$75	\$84	—	\$65
Buffalo Bills	100 level	\$86	\$108	\$94	\$79
	300 level	\$50	\$68	\$58	\$48

Our second source of data are Internet Archive's Wayback Machine search tool,⁴ which enables us to collect two factors summarizing the resale market activity for each regular season game. During the 2012–2015 seasons, StubHub was in a partnership with ESPN and provided real-time information to ESPN on the get-in price (i.e., the minimum ticket price) and the number of tickets listed for each game. We create a tool in Python that searches for the archived webpages of the ESPN website through Internet Archive's Wayback Machine and capture 48,371 snapshots (i.e., pairs of these two records for a game at a given time) to examine customers' resale market activities. A detailed description of the extraction process is given in Online Appendix A.

We also use multiple resources to obtain additional team- and demographics-related information to use as controls and/or balance the customer markets of teams that we analyze. We capture their seating maps and ticket prices using the snapshots of team websites accessed through the Internet Archive's Wayback Machine search tool. Data on their average ticket prices come from the annual reports of Team Marketing Report.⁵ We use Pro-Football-Reference.com to obtain data on stadium seat capacities. We collect preseason and weekly Vegas odds for a team to win the Super Bowl, provided by the Sports-OddsHistory.com, to take the team's expected strength and real-time performance into account, respectively. We download FiveThirtyEight's NFL Elo ratings to measure the teams' strength based on the historical game-by-game results.⁶ We use NFL.com and TheSportsDaily.net to obtain the high-profile player movements.⁷ Finally, we obtain the team values prior to the 2014 season from Forbes' 17th annual NFL Team Valuations list.⁸

In addition, we collect demographic data for each team's hometown. We obtain information on income (i.e., median household income and the Gini coefficient of household income) and population from the U.S. Census Bureau's website⁹ and ethnic diversity from Brandeis University Heller School for Social and Management's DiversityData.org project.¹⁰

Our final data set is composed of all the previously mentioned records for all 32 teams in the NFL. We also review news articles on Factiva to determine whether any of the teams experience a stadium change or irregularities in ticket sales during our observation period. We find that the San Francisco 49ers start to play their games in the new Levi's Stadium in the 2014 season, the Minnesota Vikings have to move to the University of Minnesota Stadium for the 2014 season after the collapse of the Metrodome's roof due to a snowstorm, and the Tennessee Titans franchise is reported to have been using a Florida-based broker to maintain its sellout streak since 1999. Hence, we exclude the home games of these three teams from our data set. We also check whether any one of the remaining teams' home games are played outside their home stadium due to either

NFL's promotional commitment (i.e., internationally hosted regular season games) or unexpected extreme weather. We find 11 such games and exclude them from our analysis.¹¹

Table 2 presents the summary statistics of variables used in our analyses. The upper and middle panels provide pre- and posttreatment period means and standard deviations for different variables at home game, team season, and resale market snapshot levels across teams that switch to variable pricing (i.e., treatment group) and those that do not (i.e., control group). The lower panel provides the mean and standard deviations for different demographic factors through our observation period.

4. Impact of Variable Pricing on Primary Market Ticket Sales

In this section, we first present our identification strategy to estimate the causal impact of variable pricing on primary market ticket sales. We then share the estimation results for the main specification, several robustness checks, and a model with relative time effects.

4.1. Model-Free Evidence

We first examine how capacity utilization (i.e., primary market ticket sales divided by the stadium seat capacity) for home games of the teams that switch to variable pricing (i.e., treated games) and home games of those that do not (i.e., control games) change with the switch to variable pricing. Figure 1 shows the average capacity utilization for the treated and control games in our observation period. A comparison between the percentage changes (from pre- to posttreatment period) in capacity utilization for treated ($95.62\% - 92.76\% = 2.86\%$) and control games ($98.64\% - 98.19\% = 0.45\%$) reveals that the capacity utilization increases 2.41% ($p < 0.01$) more for treated games relative to control games.

4.2. Empirical Strategy: Difference-in-Differences with Propensity Score Weighting

We can use a difference-in-differences (DiD) identification strategy (Angrist and Pischke 2008) to estimate the main effects of the switch to variable pricing, because only 15 teams switched immediately after the NFL allowed variable pricing prior to the beginning of the 2014 season and only 3 more teams followed the original 15 in the 2015 season. Specifically, we estimate the effects of variable pricing on the variables of interest (i.e., capacity utilization, the minimum ticket price on the resale market, and the number of ticket listings on the resale market) by comparing the home games of teams that switch to variable pricing (i.e., treatment group) and home games of teams that do not (i.e., control group), before and after the switch to variable pricing (i.e., treatment). The teams in the treatment and control groups are listed in Table B1 in Online Appendix B.

Table 2. Descriptive Statistics and Raw Data Patterns

Time window	Level/variables	Treatment group (teams = 12; games = 374; snapshots = 17,772)		Control group (teams = 14; games = 447; snapshots = 21,714)	
		Mean	Standard deviation	Mean	Standard deviation
Pretreatment (2012–2013)	Home games				
	Primary market ticket sales	64,275.5	5,986.8	69,928.3	9,711.9
	Capacity utilization	0.9276	0.0807	0.9819	0.0685
	Weekly Super Bowl odds	0.0433	0.0744	0.0393	0.0464
	Team-seasons				
	Stadium seat capacity	69,390.9	4,030.9	71,053.0	7,166.4
	Average ticket price	74.5979	15.4159	84.4875	16.8498
	Preseason Vegas odds	0.0394	0.0394	0.0455	0.0352
	High-profile movements (net)	−0.0833	1.5581	−0.1071	2.1489
	Resale market snapshots				
	Minimum resale price	51.4124	40.7401	59.1189	40.4168
	Number of resale listings	6,199.8	3,131.4	5,366.1	2,886.1
	Days before the game	72.8095	42.2105	70.7857	41.5149
Posttreatment (2014–2015)	Home games				
	Primary market ticket sales	65,789.3	5,351.8	70,625.5	9,230.6
	Capacity utilization	0.9562	0.0572	0.9864	0.0550
	Weekly Super Bowl odds	0.0625	0.0997	0.0432	0.0683
	Team-seasons				
	Stadium seat capacity	68,827.3	3,811.3	71,441.3	7,771.2
	Average ticket price	78.1346	16.1265	90.0354	16.8098
	Preseason Vegas odds	0.0496	0.0545	0.0458	0.0406
	High-profile movements (net)	−0.1250	1.3290	0.0714	1.9231
	Resale market snapshots				
	Minimum resale price	84.8997	61.8412	89.0199	54.8461
	Number of resale listings	6,213.1	3,496.0	5,493.0	3,418.3
	Days before the game	64.7107	44.0996	64.8515	43.8659
All (2012–2015)	Teams				
	Medium income	54,396.3	9,030.5	61,279.0	12,376.0
	Gini income	0.4635	0.0168	0.4702	0.0201
	Population (millions)	2.9437	1.9485	5.1527	4.8556
	Gini ethnicity	0.4926	0.1252	0.5593	0.1342

Note. The three NFL teams that switched to variable pricing in 2015 are not included in the table to make the pre- and post-treatment periods comparable for treatment and control groups, but these teams are used in the empirical analyses.

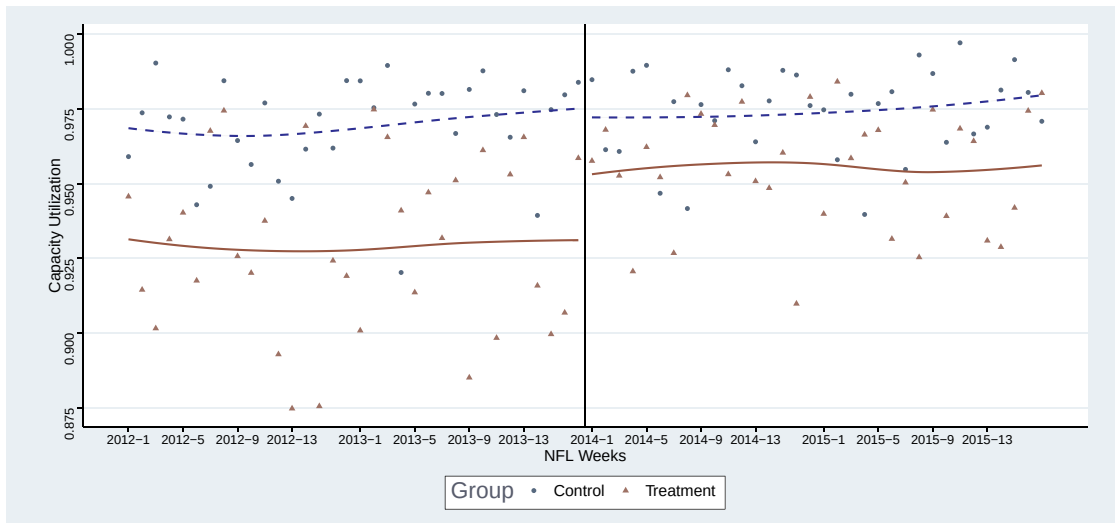
While using this identification strategy, we include the following: (i) control variables for the home team's location-specific demographic factors (i.e., median income, income diversity, population, and ethnicity diversity), pricing information (i.e., average ticket price), performance indicators (i.e., preseason and weekly odds of winning the Super Bowl), and high-profile player movements before the season; (ii) team fixed effects for unobserved home team-specific time-invariant factors (e.g., customers' preference for watching games in the stadium); (iii) opponent fixed effects for unobserved away team-specific time-invariant factors (e.g., customers' preference for supporting their team by attending away games); and (iv) NFL week and yearly fixed effects for unobserved time trends (e.g., changes in game importance).

Although the NFL's decision to allow teams to use variable pricing came just before the release of the 2014 season's game schedule, the teams' switch to variable

pricing may not be exogenous. Specifically, a team's decision to adopt variable pricing may be influenced by factors such as the demographic characteristics of the team's location (see Table C1 in Online Appendix C for the list of all factors). Therefore, we use a widely used observation weighting approach, propensity score weighting, from a family of propensity score-based methods introduced by Rosenbaum and Rubin (1983) to further reduce concerns about potential endogenous selection (see Bell et al. 2017 for a similar approach). In principle, the goal is to decrease the possibility for unobservable differences between the treated and control games by reducing the imbalances of observable characteristics between the two groups.

The propensity score is defined as the probability that a unit receives the treatment, conditional on its observed characteristics. Propensity score weighting is a method that uses propensity scores as sampling

Figure 1. (Color online) Impact of Variable Pricing on Primary Market Ticket Sales: A Graphical Analysis



Notes. The solid (dashed) line represents the fitted trend of the treatment (control) group. The three NFL teams that switch to variable pricing in 2015 are not included in the figure to make the post- and pretreatment periods comparable for treatment and control groups, but these teams are used in the empirical analyses.

weights to make the two groups comparable with respect to their observable factors (Rosenbaum 1987, Hirano and Imbens 2001, Hirano et al. 2003). This method avoids undesirable loss of subjects (Guo and Fraser 2009) and fits well to our setting due to the small number of the NFL teams. Following Hirano and Imbens (2001), we define the sampling weights as

$$\omega(T, x) = \frac{T}{\hat{\pi}(x)} + \frac{1-T}{1-\hat{\pi}(x)},$$

where T indicates a team's home game being treated and $\hat{\pi}(x)$ is the estimated probability of being treated based on the observable factors. We calculate these weights at the team level. We then use these propensity score weights in the weighted least squares estimation, which minimizes the weighted sum of the squared residuals, following Abadie (2005). In our propensity score weighting, we use the following: (i) the demographic factors of teams' hometowns (i.e., median income, Gini coefficient for income, population, and Gini coefficient for ethnicity); (ii) the mean and standard deviation of capacity utilization, and average ticket prices for the 2013 season; and (iii) stadium capacity, team value, the preseason Super Bowl odds, high-profile player movements, and the mean and standard deviation of opponent Elo ratings, which could all be determinants of a team's decision to switch to variable pricing in 2014. We use logistic regression to calculate the propensity scores (see Table C1 in Online Appendix C for the propensity score estimation results).

Following Guo and Fraser (2009), we compare the characteristics of the two groups based on the averages of the weighted characteristics. Table C2 in Online Appendix C shows the averages for each factor in the treatment and

control groups and the resulting t tests of their comparisons. These tests show no statistical evidence to reject that the two groups have the same averages at the 10% significance level and suggest that the games in the treatment and control groups are comparable in terms of our key variables.

4.3. Effect of the Switch to Variable Pricing on Primary Market Ticket Sales

We next estimate the average effect of the switch to variable pricing on primary market ticket sales. Given the differences in stadium seat capacities across teams, we use the capacity utilization ($CapUtil$, defined by the primary market ticket sales divided by the stadium seat capacity) as our dependent variable for the primary market ticket sales throughout the paper. Our unit of analysis is at the team-home game level. We identify the effect of variable pricing using

$$CapUtil_{ig} = \alpha_i + \beta VP_{ig} + W_{ig} + Y_{ig} + O_{ig} + X'_{ig}\gamma + \epsilon_{ig}, \quad (1)$$

where the indicator variable VP_{ig} is one if team i has already switched to variable pricing during its home game g and zero otherwise. Team fixed effects, α_i , capture the time-invariant unobserved factors of each team. The set of indicator variables W_{ig} represents 17 different NFL regular season weeks for team i 's home game g . The indicator variable representing the week in which a team i 's home game g takes place is one, and all other indicator variables are zero. For instance, if a team's first home game is in the second NFL week, the corresponding dummy in the vector for the second NFL week will take

the value of 1, whereas others take zero. Y_{ig} is the set of indicator variables that represent the four seasons in 2012–2015. The indicator variable representing the season in which team i 's home game g takes place is one, and all other indicator variables are zero. O_{ig} is the set of indicator variables that represent opponent team fixed effects. The vector X_{ig} consists of team- and game-related control variables. It includes a team's hometown demographic factors: median income, the natural logarithm of the Gini coefficient of income, population, and the natural logarithm of the Gini coefficient of ethnicity. X_{ig} also includes the yearly average ticket price within the season, the percentage change in the average ticket price compared with the previous season, preseason Vegas odds for the team to win the Super Bowl, and the net difference in high-profile player movements (i.e., the number of high-profile players joined minus the number of those that left). We also include teams' corresponding weekly Vegas odds of winning the Super Bowl in X_{ig} . Finally, ϵ_{ig} is the error term. Hereafter, we use the same α_i , W_{ig} , Y_{ig} , O_{ig} , and X_{ig} terms for each DiD specification and introduce superscripts when necessary. Our coefficient of interest in Equation (1) is β . Its estimated value gives us the average effect of the switch to variable pricing on the capacity utilization.

4.3.1. Identification Check. We perform an identification check to examine whether our empirical strategy can extract the causal effect of the switch to variable pricing prior to the estimation of Equation (1). The smooth trend lines in Figure 1 (obtained using a locally weighted smoothing approach) fitted to each group's observations provide strong visual evidence for parallel trends prior to the 2014 season. To validate our parallel trend observation, we also run a set of parallel trend tests. We report the results of these tests in Online Appendix D. To strengthen the causal interpretation of our main result, we also perform a placebo test similar to that performed in Nicolae et al. (2017). We report the details of this test also in Online Appendix D.

4.3.2. Main Effect on Primary Market Ticket Sales. Column (1) of Table 3 presents the estimated effect of the switch from fixed to variable pricing on capacity utilization using Equation (1) with propensity score weighting. It shows that the average effect of the switch to variable pricing on capacity utilization is positive and statistically significant ($\hat{\beta} = 0.0137$, $p < 0.05$).

Because NFL teams may choose to increase or decrease their average ticket price between seasons, we also add an interaction term between VP and the percentage change in average ticket price across seasons ($\% \Delta ATP$) to Equation (1). Column (2) of Table 3 provides the coefficient estimates for this analysis. The average effect of the switch to variable pricing on capacity utilization is still positive and statistically significant ($\hat{\beta} = 0.0159$, $p < 0.01$). This effect amounts to a

Table 3. Impact of Variable Pricing on Primary Market Ticket Sales

Variables	(1)	(2)	(3)	(4)
VP	0.0137* (0.0057)	0.0159** (0.0061)	0.0162** (0.0052)	0.0197*** (0.0058)
$VP \times \% \Delta ATP$	—	−0.0007 (0.0007)	—	−0.0013 (0.0007)
Team fixed effects	Yes	Yes	Yes	Yes
Game week fixed effects	Yes	Yes	Yes	Yes
Season fixed effects	Yes	Yes	Yes	Yes
Opponent fixed effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	917	917	917	917
PSW	Yes	Yes	No	No
R^2	0.6274	0.6278	0.6627	0.6642

Notes. Robust standard errors are in parentheses. VP , indicator of switching to variable pricing; $VP \times \% \Delta ATP$, interaction between variable pricing and change in the average ticket price; PSW, propensity score weighting.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

1.59% increase in capacity utilization in response to a switch to variable pricing. The coefficient of the interaction term is -0.0007 , but not statistically significant ($p > 0.10$). Finally, we estimate these two models without using propensity score weighting and share the results in columns (3) and (4) of Table 3.

4.4. Robustness Analyses

We perform several robustness checks to validate the direction and significance of our findings.

4.4.1. Robustness to Propensity Score Weighting Approaches. Three teams (i.e., the Cleveland Browns, the Denver Broncos, and the New York Jets) switch to variable pricing in the 2015 season, a season after the original 15 teams. To check the robustness of our propensity score weighting approach, we first re-estimate the specification in Equation (1) under alternative scenarios excluding some portions of the data and recalculating the propensity score weights when necessary: First, we exclude the 2015 season home games of teams that switch to variable pricing in the 2015 season and place these teams into the control group because they do not use variable pricing before 2015. Second, we exclude all home games (2012–2015) of teams that switch to variable pricing in the 2015 season. Third, we keep the teams that switch to variable pricing in the 2015 season in the treatment group but exclude their 2014 season home games where they do not use variable pricing. Columns (1)–(3) of Table 4 present the results, respectively. In all these alternatives, we continue to find that the impact on capacity utilization due to the switch from fixed to variable pricing is positive and statistically significant.

Inspired by the recent discussion of multiple approaches that focuses on aggregating group-time average

Table 4. Robustness to Propensity Score Weighting Approaches

Variables	(1)	(2)	(3)	(4)
VP	0.0232** (0.0071)	0.0230** (0.0073)	0.0198** (0.0069)	0.0201*** (0.0075)
Observations	893	821	893	917

Notes. Robust standard errors are in parentheses. VP, indicator of switching to variable pricing. In these specifications, team, game week, season, opponent fixed effects, controls, and $VP \times \% \Delta ATP$ are also included.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

treatment effects without data exclusion (Callaway and Sant’Anna 2021), we identify the team-year treatment effects separately (i.e., the effect of variable pricing in 2014 for those teams that switch in 2014, the effect of variable pricing in 2015 for those teams that switch in 2014, and the effect of variable pricing in 2015 for those teams that switch in 2015) using the corresponding propensity score weights and aggregate them based on the weighted average approach shown in equation (3.1) of Callaway and Sant’Anna (2021). Column (4) of Table 4 reports the result for this analysis. The group-time average treatment effect is positive and statistically significant.

4.4.2. Bootstrapping. In our analysis, a potential concern may be that one or a few teams drive the treatment effect. To check the robustness of our results against this issue, we use four different bootstrapping techniques¹² and report our results in columns (1)–(4) of Table 5. First, we use the classic Jackknife technique (i.e., systematically leaving out each game from the data, calculating the estimate, and then finding the average of these calculations as discussed in Wu (1986)). Second, we use a Jackknife technique where we leave out all the home games of a team from the data. Third, we use a similar technique as with the second, but leave out all the home games of a treated team from the data. Fourth, we use the classic Bootstrapping technique (i.e., random sampling with replacement) discussed in Efron (1981) with 1,000 replications. The impact of a switch to variable

Table 5. Robustness to Potential Outliers

Variables	(1)	(2)	(3)	(4)
VP	0.0159*** (0.00021)	0.0159*** (0.00333)	0.0157*** (0.00391)	0.01601*** (0.0062)
Bootstrap	JK-917	JK-29	JK-Treated	B-1000

Notes. Robust standard errors are in parentheses. VP, indicator of switching to variable pricing. In these specifications, team, game week, season, opponent fixed effects, controls, and $VP \times \% \Delta ATP$ are also included.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 6. Robustness to Alternative Estimation Approaches and Dependent Variables

Variables	(1)	(2)	(3)	(4)
VP	0.0203** (0.0070)	0.0214** (0.0068)	1353.27** (427.47)	0.0210*** (0.0060)
Estimation approach	Tobit	Weighted LS	Weighted LS	Poisson R
Dependent variable	CapUtil	$\ln(PMSales)$	PMSales	PMSales
Observations	917	847	847	847

Notes. Robust standard errors are in parentheses. VP, indicator of switching to variable pricing. In these specifications, team, game week, season, opponent fixed effects, controls, and $VP \times \% \Delta ATP$ are also included.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

pricing on the capacity utilization remains positive and statistically significant with each of these techniques.

4.4.3. Robustness to Estimation Approach and Alternative Measures. As mentioned earlier, we use *CapUtil* as our dependent variable due to differences in stadium seat capacities across teams. Because this variable is bounded from above at 100% (i.e., sell out), we use the Tobit model (Wooldridge 2010) as an alternative estimation approach. Column (1) of Table 6 reports the result of this analysis. The treatment effect is still positive and statistically significant.

We also check the robustness of our results to different functional specifications of primary market ticket sales. In place of *CapUtil*, we use two alternative measures: natural logarithm of primary market ticket sales ($\ln(PMSales)$), and primary market ticket sales (*PMSales*). While using these measures in our estimations, we exclude home games in seasons following a stadium renovation¹³ and include the stadium seat capacity in the set of control variables. Columns (2) and (3) of Table 6 present these results. Because our primary market ticket sales variable can also be viewed as a count variable, we also estimate a Poisson regression model (Wooldridge 2010) and report the results in column (4) of Table 6. All these robustness checks show that the switch to variable pricing has a positive and statistically significant impact on primary market ticket sales.

4.4.4. Test for Alternative Explanations. All NFL teams except for the New Orleans Saints eventually started implementing variable pricing. Having only one remaining team may indicate that some unobservable differences have prevented this team from switching to variable pricing. To address potential biases that may arise from this one control team, we re-estimate our specifications in columns (1) and (2) of Table 3 by excluding the New Orleans Saints’ home games from our sample. Table 7 shows the estimation results. Our

Table 7. Impact of Variable Pricing on Primary Market Ticket Sales (Excluding the New Orleans Saints)

Variables	(1)	(2)
VP	0.0130* (0.0060)	0.0160* (0.0063)
VP × %ΔATP	—	−0.0009 (0.0008)
Observations	885	885

Notes. Robust standard errors are in parentheses. VP, indicator of switching to variable pricing. In these specifications, team, game week, season, opponent fixed effects, and controls are also included.

* $p < 0.05$.

previous findings stay robust to the exclusion of the New Orleans Saints from our sample.

4.4.5. Potential Confounding Factors. We also examine other potential changes around the time NFL teams switch to variable pricing that might confound the relationship between ticket sales in the primary market and a switch to variable pricing. In particular, we look for a change in teams' stadium seat capacity, average ticket price, and player movements around the time they switch to variable pricing.

Using the approach suggested in Cui et al. (2020), we perform a DiD analysis similar to Equation (1) using stadium seat capacity, average ticket prices, and net difference in high-profile player movements as dependent variables. Table 8 presents the results for these analyses. The effect of the switch to variable pricing on these three variables is not significant. This analysis rules out potential confounding factors that could bias our estimates.

4.5. Relative Time Effects

We next study how the estimated effect may be different based on the length of exposure to the treatment. To do so, we implement relative-time models (see Dhanorkar 2019 and Cui et al. 2019 for two recent examples). In addition to providing a sense of long-run impact of the variable pricing (e.g., whether it weakens with time), this approach helps us to test whether the difference between treated and control games showed a trend before the introduction of variable pricing (over the

Table 8. Potential Confounding Factors

Variables	(1)	(2)	(3)
VP	−715.48 (569.97)	−0.8363 (1.6777)	−0.1598 (0.6348)
Dependent variable	Stadium seat capacity	Average ticket price	Player movements (net)
Observations	116	116	116

Notes. Robust standard errors are in parentheses. VP, indicator of switching to variable pricing. In these specifications, team, game week, season, opponent fixed effects, controls, and VP × %ΔATP are also included.

parallel trends and placebo tests performed earlier). In general, we expect no significant effects in the seasons before the switch to variable pricing, and significant and positive effects in the seasons following the switch to variable pricing.

We code relative time as the number of seasons before and after the switch to variable pricing for the home games of a treated team. This procedure allows us to compare the “treatment” across home games of teams that switch to variable pricing across seasons. We use the following equation:

$$\begin{aligned} CapUtil_{ig} = & \alpha_i + \beta_1 TwoSeasonsPrior_{ig} \\ & + \beta_2 SeasonOfSwitch_{ig} + \beta_3 OneSeasonAfter_{ig} \\ & + W_{ig} + Y_{ig} + O_{ig} + X'_{ig}\gamma + \epsilon_{ig}, \end{aligned} \quad (2)$$

where $TwoSeasonsPrior_{ig}$ is a dummy variable that is equal to one for home games that are played two seasons prior to a team's switch to variable pricing (i.e., leads). Similarly, $SeasonOfSwitch_{ig}$ is a dummy variable that is equal to one for home games that are played in the season that a team switches to variable pricing, and $OneSeasonAfter_{ig}$ is a dummy variable that is equal to one for home games that are played one season after a team switches to variable pricing (i.e., lags). The base category in this relative time model regression is the season before the switch to variable pricing. The home games of teams that never switch to variable pricing are also grouped in the base category.

We report our results in column (1) of Table 9. The estimates show no effects two seasons prior to the switch to variable pricing, with sharply increasing effects on the

Table 9. Relative Time Effects: Season and Half-Season Analyses

Variables	(1)	Variables	(2)
<i>TwoSeasonsPrior</i>	0.0019 (0.0121)	(−2.F)	0.0136 (0.0107)
<i>OneSeasonPrior</i>	—	(−2.S)	−0.0014 (0.0127)
<i>SeasonOfSwitch</i>	0.0170** (0.0064)	(−1.F)	0.0121 (0.0103)
<i>OneSeasonAfter</i>	0.0148+ (0.0083)	(−1.S)	
		(+0.F)	0.0234* (0.0107)
		(+0.S)	0.0233* (0.0092)
		(+1.F)	0.0200 (0.0122)
		(+1.S)	0.0212+ (0.0114)
Observations	917	Observations	917
R ²	0.6278	R ²	0.6294

Notes. Robust standard errors are in parentheses. In these specifications, team, game week, season, opponent fixed effects, controls, and VP × %ΔATP are also included.

+ $p < 0.10$; * $p < 0.05$; ** $p < 0.01$.

capacity utilization in the season of the switch, which then appear to flatten out following the season of switch. The left panel of Figure 2 confirms this trend.

We perform the same analysis also in the half-season level by dividing the seasons into two halves. These halves are coded as $(-2.F)$, $(-2.S)$, $(-1.F)$, $(-1.S)$, $(+0.F)$, $(+0.S)$, $(+1.F)$, and $(+1.S)$, where the sign is the indicator for before and after the switch to variable pricing ($-$ and $+$, respectively), the number presents the relative season, and the letter indicates the half of the season (F for the first and S for the second half). In this regression, the second half of the season before the switch to variable pricing (i.e., $-1.S$) is the base category. Column (2) of Table 9 and the right panel of Figure 2 present the coefficients of this analysis. We observe a similar pattern to the results of the season level analysis.¹⁴

5. Variable Pricing and Customer Decision Making in Primary and Resale Markets

In Section 4, we show that a switch to variable pricing has a positive impact on primary market ticket sales. In this section, we explore the mechanism behind this positive impact. To do so, we first present a simple economic model that captures the main features of the interplay between the primary and resale markets and explains how the changes in the primary market prices through variable pricing may affect customers' decision making. We next empirically examine the changes in the primary market ticket sales for games that became cheaper and those that became more expensive, relative to previous prices, and provide evidence that customers have an overall favorable reaction to variable pricing. Finally, we examine the impact of variable pricing

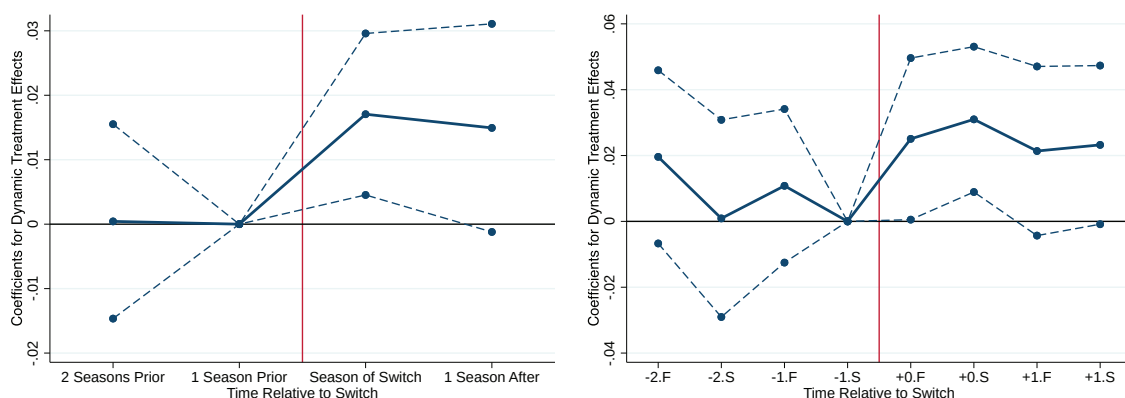
on the resale market and link the theory and our empirical observations.

5.1. Model

A switch from fixed to variable pricing enables sports teams to offer different prices in the primary market for different games instead of offering the same price for each game. Intuitively, one would expect that the primary market ticket sales for a game that is now priced lower (relative to previous seasons) increase, whereas the primary market ticket sales for a game that is now priced higher (relative to previous seasons) decrease, following the classical Law of Demand (Marshall 1892). However, when there is a resale market, customers may recover a substantial portion of their investment or actually make money by reselling a ticket. This value created by the resale market is called the option value (Bennett et al. 2015), and it can generate interesting dynamics for customers' actions. To highlight these dynamics, we study the decision making of customers in the presence of a resale market with an economic model as follows.

5.1.1. Ticket Market Characteristics. Consider a ticket market for a single sports game. The ticket market consists of a primary market and a resale market. We assume that the ticket sales for this game proceed in two subsequent periods ($T = \{1, 2\}$) prior to the game. In the first period ($T = 1$), all customers arrive to the primary market, and each can purchase a ticket at price p_1 . In the second period ($T = 2$), some of the tickets purchased from the primary market are listed on the resale market at price p_2 , and the remaining customers can purchase these tickets from the resale market. We assume that the resale market ticket price (p_2) can either be low (i.e., $p_{2l} < p_1$) or high (i.e., $p_{2h} > p_1$), but its exact value is not known in the first period.

Figure 2. (Color online) Coefficients from the Relative Time Models: Season and Half-Season Analyses



Notes. The solid line shows the coefficient estimates. The dashed lines are 95% confidence intervals. (Left) Season level analysis. For teams that switch to variable pricing in 2015, the games in 2012 are coded as *TwoSeasonsPrior*. (Right) Half-season level analysis. For teams that switch to variable pricing in 2015, the games in 2012 are coded as $(-2.F)$.

5.1.2. Customer Characteristics. Each customer is interested in a single ticket for the game. Customers are heterogeneous in their valuations for attending the game, which is characterized by their type $v \in [0, 1]$. Prior to a purchase decision in the first period, each customer has an anticipation on the likelihood that the resale market price will be low, characterized by $\phi \in [0, 1]$. We further assume that customers are heterogeneous with respect to ϕ .

5.1.3. Customer Decision Making. In the first period, customers decide whether to purchase a ticket from the primary market to maximize their expected net gain. In the second period, those customers who purchased tickets earlier decide whether to keep their tickets or sell them on the resale market at the realized resale market listing price. On the other hand, the customers who delayed their purchases to the second period decide whether to purchase the tickets at the realized resale market listing price. Consistent with the practice in major sports leagues, we assume that there exist some brokers who experience smaller transaction costs due to their familiarity with the resale markets (Cui et al. 2014) and have easier access to unsold primary market tickets (Courty 2003). We assume that if any unsold tickets are left from the first period and a high resale market price is realized in the second period, these brokers will immediately pull these remaining tickets from the primary market and make them available to the remaining customers at p_{2h} . If a low resale market price is realized in the second period, the remaining customers will obviously prefer to purchase the ticket at p_{2l} from the resale market rather than at p_1 from the primary market. We next solve this model to demonstrate the customers' decisions in the first and second periods.

Proposition 1. *There exists a threshold $\phi^*(p_1, p_{2l}, p_{2h}) \doteq (p_{2h} - p_1) / (p_{2h} - p_{2l})$ such that all customers with $\phi \leq \phi^*(.)$ purchase tickets from the primary market in the first period and the remaining customers wait for the resale market. Furthermore, this threshold $\phi^*(.)$ decreases in p_1 .*

Proposition 1 states that for a customer with a given ϕ , there is a unique decision in the first period.¹⁵ If a customer's anticipated likelihood of low resale market price is low (i.e., $\phi \leq \phi^*(.)$), the customer purchases the ticket from the primary market in the first period at p_1 to avoid paying an excessively high price in the second period or to make money through reselling. On the other hand, if a customer anticipates a higher likelihood for the low resale market price (i.e., $\phi > \phi^*(.)$), the customer chooses to wait for the resale market to avoid paying an unnecessarily high price in the first period. Consequently, if the team increases (decreases) the price of the tickets for a game with variable pricing, there will

be fewer (more) people interested in purchasing tickets from the primary market (i.e., $\partial \phi^*(.) / \partial p_1 < 0$) for given anticipated resale market prices.

Proposition 2. *Among the customers who purchase tickets from the primary market in the first period, (i) those with $v < p_{2l}$ list their tickets on the resale market, (ii) those with $v > p_{2h}$ keep their tickets, and (iii) those with $p_{2l} \leq v \leq p_{2h}$ keep their tickets if the realized resale price is p_{2l} , and list their tickets on the resale market otherwise. Among the customers who wait for the resale market, (iv) those with $v < p_{2l}$ do not purchase tickets, (v) those with $v > p_{2h}$ purchase tickets from the resale market, and (vi) those with $p_{2l} \leq v \leq p_{2h}$ purchase tickets from the resale market if the realized resale price is p_{2l} , and do not purchase tickets otherwise.*

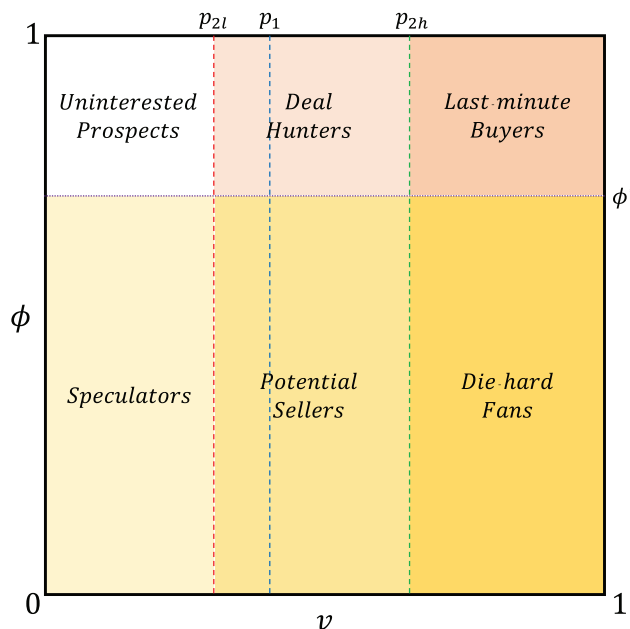
Proposition 2 suggests six customer segments based on customers' first and second period decisions. We label the customers in these segments accordingly with their observed decisions in the first and second periods. The first three segments purchase tickets from the primary market. The first segment consists of *speculators*, who are not interested in going to the game (i.e., $v < p_{2l}$) but see an arbitrage opportunity through the resale market because of high anticipation of the higher resale market price (i.e., $\phi \leq \phi^*(.)$). The second segment consists of *potential sellers*, who go to the game if p_{2l} is observed but would sell their tickets if p_{2h} is observed. This segment has two types of customers: (a) those who normally would not go to the game if the resale market did not exist (i.e., $p_{2l} \leq v < p_1$) but now take the risk for the negative utility in exchange for the arbitrage opportunity due to high anticipation of the higher resale market price (i.e., $\phi \leq \phi^*(.)$), and (b) those who would normally go to game without a resale market (i.e., $p_1 \leq v \leq p_{2h}$) but prefer the primary market over the resale market (i.e., $\phi \leq \phi^*(.)$) and now enjoy the profit if the resale market price realized is p_{2h} . The third segment is composed of *die-hard fans*, who would always go to the game regardless of the resale market price observed ($v > p_{2h}$) but prefer the primary market over the resale market ($\phi \leq \phi^*(.)$).

The remaining three segments are interested in purchasing tickets only from the resale market. The first of these segments has *uninterested prospects*, who are not interested in going to the game (i.e., $v < p_{2l}$) and do not see a potential arbitrage opportunity because of high anticipation of the lower resale market price (i.e., $\phi > \phi^*(.)$). The second one consists of *deal hunters*, who would wait for the resale market (i.e., $\phi > \phi^*(.)$) and purchase from the resale market only if p_{2l} is observed (i.e., $p_{2l} \leq v \leq p_{2h}$). The final segment is composed of *last-minute buyers*, who wait for the resale market (i.e., $\phi > \phi^*(.)$) and purchase from this market (i.e., $v > p_{2h}$). Figure 3 shows these customer segments with their

labels. This segmentation confirms how our simple economic model reflects all the possible customer decisions observed in practice (Su 2010, Cui et al. 2014).

5.1.4. Effect of Variable Pricing on Primary Market Ticket Sales. Proposition 1 suggests that a decrease (an increase) in the primary market price p_1 would lead to more (less) ticket sales in the primary market. However, this standard result is based on the assumption that no relationship between the primary market price and the anticipated resale market prices exists. After the switch to variable pricing, a change in the price of a ticket may also affect customers' anticipation of potential resale market prices (i.e., a change in p_{2l} and p_{2h}). Because posted prices represent a reference point (Kahneman et al. 1986) and may signal which games are expected to be more attractive (Bagwell and Riordan 1991), a change in the face value of a ticket is likely to change the option value for the ticket. Hence, to properly study the impact of a change in the primary market price after the switch to variable pricing, we need to account for the reference price effects of the primary market ticket price on the resale market prices. To do so, we assume that a change in the primary market ticket price of p_1 by Δ would change the anticipated resale market prices p_{2l} and p_{2h} by $w_l\Delta$ and $w_h\Delta$, respectively. In this setting, $w_l > 0$ and $w_h > 0$ represent customers' sensitivity to the changes in the primary market price in anticipation of the resale market prices. We assume $p_{2l} + w_l\Delta < p_1 + \Delta < p_{2h} + w_h\Delta$. We next demonstrate the change in primary market ticket sales after this extension.

Figure 3. (Color online) Customer Segments Based on Valuations and Anticipated Likelihood of the Lower Resale Market Price



Proposition 3. When the changes in the primary market price p_1 influence the anticipated resale market prices p_{2l} and p_{2h} , the primary market ticket sales only increase (i) after a decrease in the primary market ticket price if and only if $w_l(p_{2h} - p_1) + w_h(p_1 - p_{2l}) < p_{2h} - p_{2l}$ and (ii) after an increase in the primary market ticket price if and only if $w_l(p_{2h} - p_1) + w_h(p_1 - p_{2l}) > p_{2h} - p_{2l}$.

Proposition 3 shows that a change in the option value through changes in the anticipated resale market price competes against the direct effect of the primary market price and may even dominate it in several scenarios: For those games that became cheaper (i.e., $\Delta < 0$), the effect of the price change on the primary market ticket sales is only positive if the customers expect the resale market prices not to decrease drastically for these games (i.e., relatively lower w_l and w_h). On the other hand, for those games that became more expensive (i.e., $\Delta > 0$), if the customers expect the resale market prices to increase drastically (i.e., relatively higher w_l and w_h), it is possible that the primary market ticket sales increase. Hence, the impact of a price change through variable pricing on the primary market ticket sales is unclear in the presence of resale markets. To shed light on our estimated effect in Section 4, we explore the changes in primary market ticket sales and the prices (and the number of ticket listings) on the resale market for those games that became cheaper and more expensive separately in Sections 5.2 and 5.3.

5.2. Customers' Response to Variable Pricing

The analysis presented in Section 4 focuses on the average effect of the switch to variable pricing. However, our simple economic model in Section 5.1 shows that the effect can be positive only in certain scenarios and may differ for those games that became cheaper versus those that became more expensive.

Therefore, we first check how customers reacted to lower prices for the less attractive games or higher prices for the more attractive games after the switch to variable pricing. To explore how the impact of variable pricing changes based on the attractiveness of a game, we first need to differentiate between the less and more attractive games. However, this is not a trivial task, because all teams in the 2012 and 2013 seasons, 17 of 32 teams in the 2014 season and 14 of 32 teams in the 2015 season use a fixed pricing strategy. One solution would be to consider the game attendance; however, this may be problematic due to potential real-time effects. For example, a bad performance by the team or bad weather may constrain the attendance numbers for a game perceived as more attractive at the beginning of the season.

To solve this issue, we build a machine learning technique using historical attendance and detailed NFL team and game data to identify *unattractive* and *attractive*

games in the 2012 and 2013 seasons and the 2014 and 2015 seasons for teams that do not switch to variable pricing. Our test run using the 2014 data of teams that switch to variable pricing show that our algorithm correctly classifies 85.4% of the home games selected as high-priced games by these teams. We provide details of our approach in Online Appendix F.

We define an indicator variable, Att_{ig} , to be one for the games that our algorithm identified as attractive in the 2012–2015 seasons when a game's hometown team does not use variable pricing, or for the high-priced games of teams that switch to variable pricing and zero otherwise. To assess the differing effect of the switch to variable pricing based on the attractiveness of the game, we estimate the following specification:

$$CapUtil_{ig} = \alpha_i + \beta VP_{ig} + \theta VP_{ig} \times \% \Delta ATP_{ig} + \eta VP_{ig} \times Att_{ig} + W_{ig} + Y_{ig} + O_{ig} + X'_{ig} \gamma + \epsilon_{ig}, \quad (3)$$

where η gives the heterogeneous impact of the switch to variable pricing on capacity utilization in terms of the game attractiveness. Column (1) in Table 10 provides the coefficient estimates for this specification. The coefficient estimate of the VP_{ig} is positive and statistically significant ($\hat{\beta} = 0.0168$, $p < 0.05$) and the coefficient estimate of the $VP_{ig} \times Att_{ig}$ is not statistically significant ($\hat{\eta} = -0.0026$, $p > 0.10$). These estimates show that (i) lower prices for less attractive games are well received by the customers, and (ii) customers do not have an unfavorable reaction to higher prices for more attractive games.¹⁶ In other words, the positive effect of variable pricing does not change based on the attractiveness of a game. Proposition 3 suggests that this is possible if customers' sensitivity to the price changes is low for less attractive games and high for more attractive games.

Table 10. Responses to Price Changes

Variables	(1)	(2)	(3)
VP	0.0168* (0.0068)	—	—
VP × Att	−0.0026 (0.0062)	—	—
Markdown	—	0.1422* (0.0598)	—
Markup	—	—	0.0617* (0.0291)
Games	All	Unattractive	Attractive
Observations	917	516	311
R ²	0.6291	0.6728	0.7449

Notes. Robust standard errors are in parentheses. *VP*, indicator of switching to variable pricing; *Markdown*, average percentage markdown for an unattractive game; *Markup*, average percentage markup for an attractive game. In these specifications, team, game week, season, opponent fixed effects, controls, and $VP \times \% \Delta ATP$ are also included.

* $p < 0.05$.

We next examine how customers respond to actual price changes. To do this, we first divide the games into two buckets, unattractive and attractive, and calculate the game-level price changes for teams that switch to variable pricing. Because detailed pricing information is not available for all teams (e.g., Dallas Cowboys requires customers to register and log in to their online platform to see the prices), we only use games with publicly posted prices (817 of 917 games). We calculate the average markdown for unattractive games (i.e., *Markdown*) and markup for attractive games (i.e., *Markup*), with respect to average ticket price.¹⁷ For example, if the average price for an unattractive game is \$80, whereas the average ticket price across games throughout the season is \$100, the markdown is defined as 20%.

We update Equation (1) by replacing *VP* with *Markdown* and *Markup* for unattractive and attractive games, respectively:

$$CapUtil_{ig} = \alpha_i + \beta_{md} Markdown_{ig} + \theta VP_{ig} \times \% \Delta ATP_{ig} + W_{ig} + Y_{ig} + O_{ig} + X'_{ig} \gamma + \epsilon_{ig}, \quad (4)$$

$$CapUtil_{ig} = \alpha_i + \beta_{mu} Markup_{ig} + \theta VP_{ig} \times \% \Delta ATP_{ig} + W_{ig} + Y_{ig} + O_{ig} + X'_{ig} \gamma + \epsilon_{ig}, \quad (5)$$

where β_{md} and β_{mu} give the impact of the actual price change through variable pricing. Columns (2) and (3) in Table 10 provide the coefficient estimates for these alternative specifications. The coefficient estimate for *Markdown* is positive and statistically significant ($\hat{\beta}_{md} = 0.1422$, $p < 0.05$) and suggests that a price decrease for a less attractive game increases the capacity utilization. The coefficient estimate for *Markup* is also positive and statistically significant ($\hat{\beta}_{mu} = 0.0617$, $p < 0.05$) and reveals an interesting phenomenon: A price increase for a more attractive game actually increases the capacity utilization. This result suggests that option value effect compensates the negative effect of price increase on the primary market ticket sales (i.e., relatively high w_l and w_h) for more attractive games. Customers continue to purchase tickets for more attractive games in advance knowing that they may be able to sell their tickets at a higher price later. We next check this conjecture by analyzing the customers' choice of prices in the resale market.

5.3. Impact of Variable Pricing on the Resale Market Activities

We examine the impact of teams' switch to variable pricing on ticket owners' activity in the resale markets by studying the changes in minimum ticket prices and the number of ticket listings per game on the resale market. Because season and single-game ticket buyers act differently in their purchases (Tereyağoglu et al. 2017, Arslan et al. 2022) and such differences can lead to

Table 11. Impact of Variable Pricing on the Minimum Ticket Prices and the Number of Ticket Listings on the Resale Market Before the Start of the Season

Variables	MinResalePrice (1)	MinResalePrice (2)	ln(RMListings) (3)	ln(RMListings) (4)
VP	−3.3196 (1.7652)	13.7887*** (3.0220)	0.0652*** (0.0162)	0.0125 (0.0195)
Week-to-Season	Yes	Yes	Yes	Yes
Primary Market Minimum Price	Yes	Yes	No	No
Games	Unattractive	Attractive	Unattractive	Attractive
Observations	3,822	2,491	4,937	3,155
R ²	0.8059	0.8109	0.8209	0.8654

Notes. Robust standard errors are in parentheses. VP, indicator of switching to variable pricing. The dependent variable for columns (1) and (2) is the minimum ticket price on the resale market. The dependent variable for columns (3) and (4) is the natural logarithm of the number of ticket listings on the resale market. In these specifications, team, game week, season, opponent fixed effects, and controls are also included.

*** $p < 0.001$.

variation in ticket owners' resale market activities, we perform this examination for two time windows.

We first explore the resale market activities in the 21-day period before the start of the season, right after the delivery of season tickets in August. To assess the impact of variable pricing on the resale market activities for less attractive and more attractive games, we divide games into two buckets and use the following specifications for minimum ticket prices and the number of ticket listings, respectively¹⁸:

$$\begin{aligned} \text{MinResalePrice}_{igd} = & \alpha_i^{\text{MinResalePrice}} + \beta^{\text{MinResalePrice}} \text{VP}_{ig} \\ & + W_{ig} + Y_{ig} + D_d + O_{ig} \\ & + X'_{ig} \text{MinResalePrice} \gamma^{\text{MinResalePrice}} + \epsilon_{igd}, \end{aligned} \quad (6)$$

$$\begin{aligned} \ln(\text{RMListings}_{igd}) = & \alpha_i^{\text{Listings}} + \beta^{\text{Listings}} \text{VP}_{ig} + W_{ig} + Y_{ig} \\ & + D_d + O_{ig} + X'_{ig} \gamma^{\text{Listings}} + \epsilon_{igd}, \end{aligned} \quad (7)$$

where $\text{MinResalePrice}_{igd}$ is the minimum ticket price, and RMListings_{igd} is the number of ticket listings on the resale

market d days before the start of the season for team i 's home game g . D_d is the index of the number of remaining weeks to the start of the season and captures the time-variant unobserved factors for each snapshot (i.e., week-to-season fixed effects). Following Sweeting (2012), we include each game's primary market minimum ticket price as a control in $X'_{ig} \text{MinResalePrice}$ in Equation (6). Our coefficients of interest are $\beta^{\text{MinResalePrice}}$ and β^{Listings} .

We then explore the resale market activities in the 21-day period before each game using a similar strategy where $\text{MinResalePrice}_{igd}$ and RMListings_{igd} are the minimum ticket price and number of ticket listings on the resale market d days before the game for team i 's home game g .¹⁹ Tables 11 and 12 present the estimation results for the before the start of the season and before each game, respectively.

Column (1) of Tables 11 and 12 presents the effect of variable pricing on the minimum ticket price on the resale market for unattractive games. The nonsignificant coefficient estimates of VP_{ig} ($\hat{\beta}^{\text{MinResalePrice}} = -3.3196$ and $\hat{\beta}^{\text{MinResalePrice}} = 2.8637$, both $p > 0.05$) show that the minimum ticket prices on the resale market did not change

Table 12. Impact of Variable Pricing on the Number of Ticket Listings and the Minimum Ticket Prices on the Resale Market Before Games

Variables	MinResalePrice (1)	MinResalePrice (2)	ln(RMListings) (3)	ln(RMListings) (4)
VP	2.8637 (2.8593)	21.0528*** (4.2519)	0.1669*** (0.0320)	0.0692 (0.0416)
Week-to-Game	Yes	Yes	Yes	Yes
Primary Market Minimum Price	Yes	Yes	No	No
Games	Unattractive	Attractive	Unattractive	Attractive
Observations	3,236	2,099	4,425	2,665
R ²	0.7263	0.7450	0.6088	0.6166

Notes. Robust standard errors are in parentheses. VP, indicator of switching to variable pricing. The dependent variable for columns (1) and (2) is the minimum ticket price on the resale market. The dependent variable for columns (3) and (4) is the natural logarithm of the number of ticket listings on the resale market. In these specifications, team, game week, season, opponent fixed effects, and controls are also included.

*** $p < 0.001$.

(i.e., relatively low w_l and w_h) for less attractive games after the switch to variable pricing and explains the positive effect discussed in Section 5.2 (following Proposition 3). Column (2) of Tables 11 and 12 presents estimation results for the same effect for attractive games. The coefficient estimates of VP_{ig} are positive and statistically significant ($\hat{\beta}^{MinResalePrice} = 13.7887$ and $\hat{\beta}^{MinResalePrice} = 21.0528$; both $p < 0.001$). This suggests that the minimum ticket prices on the resale market increased (i.e., relatively high w_l and w_h) for more attractive games after the switch to variable pricing and explains the positive effect discussed in Section 5.2 (following Proposition 3).

Column (3) of Tables 11 and 12 presents the effect of variable pricing on the number of ticket listings on the resale market for unattractive games. The positive and statistically significant coefficient estimates of VP_{ig} ($\hat{\beta}^{Listings} = 0.0652$ and $\hat{\beta}^{Listings} = 0.1669$; both $p < 0.001$) suggest that, after the switch to variable pricing, the number of ticket listings on the resale market increased for less attractive games. Column (4) of Tables 11 and 12 presents estimation results for the same effect for attractive games. The coefficient estimates of VP_{ig} are not statistically significant ($\hat{\beta}^{Listings} = 0.0125$ and $\hat{\beta}^{Listings} = 0.0692$; both $p > 0.05$). Hence, there is no statistical evidence to suggest a change in the number of ticket listings on the resale market for more attractive games.

5.4. Link Between the Economic Model and Empirical Findings

The empirical analysis in Section 5.2 demonstrates that customers had a favorable reaction to both lower and higher prices of the variable pricing. To explain the mechanism behind the positive impact of the variable pricing, we now use the empirical results for the resale market in Section 5.3 and link them with the economic model in Section 5.1 for less attractive and more attractive games separately.

For the less attractive games that became cheaper with variable pricing, we observe that the minimum ticket prices on the resale market did not change both before the start of the season and before each game. This shows that change in the option value was not significant for these games, and therefore the change in the primary market price was the main driver of the increase in the primary market ticket sales. In addition, the higher primary market ticket sales for these games also increased the number of ticket listings. However, we did not find evidence for a change in the cannibalization threat because the prices on the resale market did not change.

On the other hand, for the more attractive games that became more expensive with the variable pricing, we observe that the minimum ticket prices increased before the start of the season and before each game. This provides evidence for an increase in the option value

for these games. Although this may sound like additional revenue left on the table for the teams, more customers purchased tickets from the primary market for more attractive games after the price increase due to variable pricing.

6. Determinants of Variable Pricing Success

After establishing how variable pricing works, we next investigate when variable pricing is the most effective. On the one hand, the new prices should match customers' economic conditions in the team's hometown, so that the new prices are favorably perceived. On the other hand, a team needs to formulate the right pricing strategy across different games and seat sections to maximize revenues. For practitioners in the field, it is important to determine whether and when these two steps contribute to the success of variable pricing.

6.1. Market Responsiveness to Variable Pricing

We start by checking whether and how the embrace of the switch to variable pricing differs based on a team's hometown demographics. Prior literature indicates that lower-income customers have a higher likelihood of searching and paying lower prices on average (Goldman and Johansson 1978). Siegfried and Peterson (2000) find that customers who buy sporting event tickets have a median income 84% above the national median income. With the switch to variable pricing, the unattractive games are now priced lower. Thus, lower-income customers, who would not normally purchase a ticket, can now purchase lower-priced tickets from the primary market. Therefore, we expect that the impact of the switch to variable pricing will be more positive in towns where income levels are lower and income diversity is higher. To examine this conjecture, we use the following specification:

$$\begin{aligned} CapUtil_{ig} = & \alpha_i + \beta VP_{ig} + \theta VP_{ig} \times \% \Delta ATP_{ig} \\ & + \eta VP_{ig} \times Att_{ig} + \delta_1 VP_{ig} \\ & \times MedianIncome_i + \delta_2 VP_{ig} \\ & \times IncomeDiversity_i + W_{ig} + Y_{ig} \\ & + O_{ig} + X'_{ig} \gamma + \epsilon_{ig}, \end{aligned} \quad (8)$$

where the variables $MedianIncome_i$ and $IncomeDiversity_i$ are the demographic characteristics for the hometown of each team i . The parameters of interest in this specification are δ_1 and δ_2 . The former measures the heterogeneous impact of the switch to variable pricing on the capacity utilization in terms of the median income, whereas the latter measures the heterogeneous impact of the switch in terms of the income diversity.

Column (1) of Table 13 shows the coefficient estimates for the new specification. The negative and statistically significant coefficient estimate of $VP_{ig} \times MedianIncome_i$ ($\delta_1 = -0.0188$, $p < 0.001$) suggests that the increase in

Table 13. Determinants of Variable Pricing Success: Market Responsiveness and Variable Pricing Positioning

Variables	(1)	(2)	(3)	(4)
VP	0.5475*** (0.1350)	0.4633*** (0.1351)	0.5130*** (0.1389)	0.9060** (0.3211)
VP × Att	−0.0017 (0.0059)	−0.0011 (0.0059)	−0.0013 (0.0059)	0.0011 (0.0082)
VP × MedianIncome	−0.0188*** (0.0046)	−0.0138** (0.0049)	−0.0148** (0.0052)	−0.0299*** (0.0087)
VP × IncomeDiversity	0.5611*** (0.1595)	0.5068** (0.1563)	0.5472*** (0.1548)	0.9705** (0.3373)
VP × GameTiers	—	0.0258** (0.0085)	0.0279** (0.0090)	0.0297* (0.0140)
VP × SeatTiers	—	—	−0.0014 (0.0016)	−0.0005 (0.0023)
VP × VPinSTStubs	—	—	—	−0.0005 (0.0221)
Observations	917	917	917	688
R ²	0.6385	0.6415	0.6418	0.6701

Notes. Robust standard errors are in parentheses. VP, indicator of switching to variable pricing. In these specifications, team, game week, season, opponent fixed effects, controls, and VP × %ΔATP are also included.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

capacity utilization in response to variable pricing is greater in a team hometown with lower income relative to a hometown with higher income. In addition, the positive and statistically significant coefficient estimate of $VP_{ig} \times IncomeDiversity_i$ ($\hat{\delta}_2 = 0.5611$, $p < 0.001$) suggests that the increase in capacity utilization due to variable pricing is greater in a team hometown with higher income diversity relative to one with lower income diversity.²⁰

It is possible that the higher impact of variable pricing in the hometowns with lower income levels and higher income diversity may be driven by the ticket pricing decisions, for example, teams in these hometowns may have chosen higher markdowns for their less attractive games. Therefore, we again divide games into two buckets (i.e., unattractive and attractive) and test whether our finding is driven by the lower prices or the actual market response. To do so, we add two interaction terms to Equations (4) and (5):

$$CapUtil_{ig} = \alpha_i + \beta_{md} Markdown_{ig} + \delta_{md1} Markdown_{ig} \times MedianIncome_i + \delta_{md2} Markdown_{ig} \times IncomeDiversity_i + \theta VP_{ig} \times \% \Delta ATP_{ig} + W_{ig} + Y_{ig} + O_{ig} + X'_{ig} \gamma + \epsilon_{ig}, \quad (9)$$

$$CapUtil_{ig} = \alpha_i + \beta_{mu} Markup_{ig} + \delta_{mu1} Markup_{ig} \times MedianIncome_i + \delta_{mu2} Markup_{ig} \times IncomeDiversity_i + \theta VP_{ig} \times \% \Delta ATP_{ig} + W_{ig} + Y_{ig} + O_{ig} + X'_{ig} \gamma + \epsilon_{ig}, \quad (10)$$

where δ_{md1} , δ_{md2} , δ_{mu1} , and δ_{mu2} measure the heterogeneous impact of the changes in primary market prices (decrease and increase) in terms of the median income and the income diversity.

Columns (1) and (2) in Table 14 provide the coefficient estimates for these alternative specifications. The coefficient estimate of $Markdown_{ig} \times MedianIncome_i$ is negative and significant ($\hat{\delta}_{md1} = -0.3047$, $p < 0.001$) and the coefficient estimate of $Markdown_{ig} \times IncomeDiversity_i$ is positive and significant ($\hat{\delta}_{md2} = 4.0074$, $p < 0.01$). This suggests that price decreases are well received in hometowns with lower income levels and higher income diversity compared with others, supporting our previous finding of the favorable perception of the variable pricing in these hometowns. On the other hand, the coefficient estimate for $Markup_{ig} \times MedianIncome_i$ is not statistically significant ($\hat{\delta}_{mu1} = -0.0613$, $p > 0.10$) and the coefficient estimate for $Markup_{ig} \times IncomeDiversity_i$ is positive and significant ($\hat{\delta}_{mu2} = 2.2633$, $p < 0.05$). This shows that the higher prices increased the capacity utilization particularly in the hometowns with higher income diversity.

6.2. Influence of Variable Pricing Positioning

We next check how the effect of variable pricing changes based on the differences in teams' adoption of variable pricing. Studies identify several inefficiencies between the extent of price differentiation and the quality of product received by different customer types (Miravete 2002, McManus 2007). Therefore, the effect of adopting a greater number of price tiers in variable pricing of games is ambiguous. In our sample, we see that some teams adopt a variable pricing with two price tiers, whereas the others use three price tiers. To examine whether such a differentiation led to a different effect, we use the following specification:

$$CapUtil_{ig} = \alpha_i + \beta VP_{ig} + \theta VP_{ig} \times \% \Delta ATP_{ig} + \eta VP_{ig} \times Att_{ig} + \delta_1 VP_{ig} \times MedianIncome_i + \delta_2 VP_{ig} \times IncomeDiversity_i + \mu_1 VP_{ig} \times GameTiers_{ig} + W_{ig} + Y_{ig} + O_{ig} + X'_{ig} \gamma + \epsilon_{ig}, \quad (11)$$

where $GameTiers_{ig}$ takes a value of one if team i uses three different price tiers when implementing variable pricing for the season of home game g and zero otherwise. The parameter of interest in this specification is μ_1 , which measures the heterogeneous impact of the switch to variable pricing for teams that adopt three different price tiers relative to those that implement only two price tiers. Column (2) of Table 13 shows the coefficient estimates. The positive and statistically significant coefficient estimate of the $VP_{ig} \times GameTiers_{ig}$ ($\hat{\mu}_1 = 0.0258$, $p < 0.01$) suggests that the adoption of more than two price

Table 14. Determinants of Variable Pricing Success: Market Responsiveness to Changes in Price

Variables	(1)	Variables	(2)
<i>Markdown</i>	0.0094 (0.0451)	<i>Markup</i>	0.0361 (0.0274)
<i>Markdown</i> × <i>MedianIncome</i>	−0.3407*** (0.0716)	<i>Markup</i> × <i>MedianIncome</i>	−0.0613 (0.0463)
<i>Markdown</i> × <i>IncomeDiversity</i>	4.0074** (1.2720)	<i>Markup</i> × <i>IncomeDiversity</i>	2.2633* (0.9995)
Games	Unattractive	Games	Attractive
Observations	516	Observations	311
R ²	0.6928	R ²	0.7521

Notes. Robust standard errors are in parentheses. *Markdown*, average percentage markdown for an unattractive game; *Markup*, average percentage markup for an attractive game. In these specifications, team, game week, season, opponent fixed effects, controls, and $VP \times \% \Delta ATP$ are also included.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

tiers leads to higher capacity utilization. This result suggests that investment in sports analytics personnel, who could identify higher differentiation across games to set new price tiers, can bring in higher primary market ticket sales.

The NFL teams have set different prices for different seat categories, which is an example of second-degree price discrimination (see Tirole (1988) for a discussion of different kinds of price discrimination). Empirical studies show a positive return from this seat-based price differentiation with a note of decreasing returns from additional seat categories (Leslie 2004, Courty and Pagliero 2012). In fact, Wilson (1993) and Miravete (2007) point to the significant cost of adopting more complex pricing tariffs. In our context, adoption of variable pricing on top of seat-based price differentiation introduces further complexity to the teams' price differentiation strategies. Hence, the increase in ticket sales from the adoption of variable pricing may be lower for teams that have more seat tiers relative to other teams. To examine this conjecture, we use the following specification:

$$\begin{aligned}
 CapUtil_{ig} = & \alpha_i + \beta VP_{ig} + \theta VP_{ig} \times \% \Delta ATP_{ig} \\
 & + \eta VP_{ig} \times Att_{ig} + \delta_1 VP_{ig} \\
 & \times MedianIncome_i + \delta_2 VP_{ig} \\
 & \times IncomeDiversity_i + \mu_1 VP_{ig} \\
 & \times GameTiers_{ig} + \mu_2 VP_{ig} \\
 & \times SeatTiers_{ig} + W_{ig} + Y_{ig} + O_{ig} \\
 & + X'_{ig} \gamma + \epsilon_{ig},
 \end{aligned} \quad (12)$$

where $SeatTiers_{ig}$ is the number of seat tiers with different prices set by team i for home game g .²¹ In this specification, μ_2 measures the heterogeneous impact of the switch to variable pricing in terms of the number of seat tiers with different prices. Column (3) of Table 13 shows the coefficient estimates. The coefficient estimate of interest is not statistically significant ($\hat{\mu}_2 = -0.0014$, $p > 0.10$). This result suggests that the impact of variable

pricing does not decrease for teams that implemented a higher degree of seat-based price differentiation.

After an extensive search for the images of the ticket stubs of the 2014 games of those teams that adopt variable pricing, we observe differences in the display of prices on the ticket stubs in the season ticket packages. Some of these teams display different prices for different games (as expected) on the ticket stubs in the season ticket packages, whereas others print the same price (i.e., the average price per game in a season ticket package) on ticket stubs for each game even though they adopt variable pricing. Such differences in the display of pricing may affect ticket purchases (Train et al. 1987). To test this conjecture, we use the following specification:

$$\begin{aligned}
 CapUtil_{ig} = & \alpha_i + \beta VP_{ig} + \theta VP_{ig} \times \% \Delta ATP_{ig} \\
 & + \eta VP_{ig} \times Att_{ig} + \delta_1 VP_{ig} \\
 & \times MedianIncome_i + \delta_2 VP_{ig} \\
 & \times IncomeDiversity_i + \mu_1 VP_{ig} \\
 & \times GameTiers_{ig} + \mu_2 VP_{ig} \times SeatTiers_{ig} \\
 & + \mu_3 VP_{ig} \times VPInSTStubs_i + W_{ig} + Y_{ig} \\
 & + O_{ig} + X'_{ig} \gamma + \epsilon_{ig},
 \end{aligned} \quad (13)$$

where the variable $VPInSTStubs_i$ takes a value of one if team i display different prices for games on their ticket stubs in their season ticket packages and zero otherwise. The variable μ_3 measures the heterogeneous impact of the switch to variable pricing for teams that show different prices for different games relative to those that do not. Column (3) of Table 13 shows the coefficient estimates. The coefficient estimate of interest is not statistically significant ($\hat{\mu}_3 = -0.0005$, $p > 0.10$). This result demonstrates that there is no significant change in the impact of variable pricing on capacity utilization when teams display different prices for games in their season ticket packages. We believe that the customers purchasing the season ticket packages were well aware of the switch to variable pricing (and therefore

new prices). Therefore, the strategy of displaying different prices for different games did not offer any added benefits.

7. Conclusion

In an age when many sports teams are considering a switch to variable pricing from the traditional fixed pricing, it is crucial to understand *whether, how, and when* variable pricing strategies can boost primary market ticket sales. Using quasi-experimental data from the NFL, this paper takes a first step to examine the impact of variable pricing on sports ticket markets and provides several key findings with important implications for both managers and policymakers not only in sports but also in the entertainment industry.

First, we find that the adoption of variable pricing leads to a 1.59% increase in primary market ticket sales per game. Lower prices for the less attractive games are well received by customers, and customers do not have an unfavorable response to higher prices for the more attractive games. However, it is not clear whether this positive effect of variable pricing will lead to revenue benefits. The NFL does not share any revenue information at the game level; therefore, we use a hypothetical NFL team based on average team-related observations in the 2012–2013 seasons to estimate revenue figures. Suppose that this team sells an average of 67,566 tickets for less attractive games and an average of 68,859 tickets for more attractive games without variable pricing. By switching to variable pricing, the team management sets the ticket prices of five less attractive games at 12% cheaper than the fixed price and those of three more attractive games at 20% more expensive than the fixed price (therefore, keeping the average ticket price the same). Using the specification in Equation (3), we find the change in ticket sales for each game and calculate the revenues with and without variable pricing. This analysis shows that the revenue impact of a switch to variable pricing is 1.71%. Because the NFL has around \$5.5 billion of stadium revenue (i.e., the sum of tickets, concessions, sponsors, parking, and team stores) per year (Ozanian 2020), the magnitude of this impact is economically significant.

Second, our analysis of the customers' actions in the resale market reveals some important insights. For the less attractive games, the minimum ticket prices do not change, and the number of ticket listings increase on the resale market. For the more attractive games, the minimum ticket prices increase, and the number of ticket listings do not change on the resale market. These findings answer two important questions that sports teams may have about the long-term impact of variable pricing. How does variable pricing affect the cannibalization of primary market ticket sales? How does variable pricing affect the scalping behavior in the primary market? For the former, although an increase in the number of the

ticket listings on the resale market for the less attractive games can lead to a higher level of cannibalization of primary market ticket sales, the minimum ticket prices do not go down for these games; therefore, we show evidence that variable pricing does not have a negative impact through increased cannibalization. For the latter, our findings for the more attractive games suggest that at least some people purchase tickets to sell later at a higher price, but the number of ticket listings on the resale market do not change for these games after variable pricing. Therefore, variable pricing does not have a negative impact through encouraging scalping in the primary market. The relative time analysis further shows that the effect of switch to variable pricing does not weaken with time and confirms that variable pricing does not have any negative consequences in the long term.

Third, we examine when variable pricing is likely to be the most effective. We first find that variable pricing is more effective in team hometowns with lower income and higher income diversity. We repeat the same revenue analysis for teams with a half standard deviation higher and lower median income level and income heterogeneity values than their means. Our analysis shows that the revenue benefits of variable pricing can be up to 3.71% for a hypothetical NFL team with a hometown that has a lower median income level and a higher income heterogeneity. This implies that it is essential to take demographics into account while making pricing decisions. Furthermore, our finding that variable pricing increases the primary market ticket sales more for team hometowns with lower income and higher income diversity supports the criticism that traditional fixed pricing strategies favor the customers with higher income. Therefore, policymakers could encourage variable pricing for organizations as a way to address this fairness issue in markets. We also find that teams with a greater price differentiation across games attain higher ticket sales. This implies that improving their sports analytics capabilities can help teams identify higher differentiation and get closer to offering the right price points.

As in any study relying on observational data, there is room for further assessment of our results. Although we have extensive information on the adoption of variable pricing, information on how this strategy is implemented for different seat categories is limited. If one can find information on sales and price for each section, our empirical models can be used to analyze the heterogeneous effect of variable pricing across different seat categories. Moreover, with individual-level transaction data for primary and resale market activity, one could disentangle the option value and demand-decreasing effect of price increases for more attractive games. We also hope that future studies will examine the effect of variable pricing as organizations in different contexts implement this pricing strategy, as we were only able to

focus on a single sports league due to the extensive data collection required. More research on the effectiveness of other types of pricing strategies would enhance our understanding of which pricing strategies work best in settings with both primary and resale markets. Finally, a structural analysis of the customers' utilities from different ticket choices and welfare implications of variable pricing are other potential directions for future research.

Acknowledgments

The authors thank Gabriel Weintraub, the associate editor, and two reviewers for thoughtful suggestions. The authors also thank Robert Clark, Mark Ferguson, Emily Garbinsky, Pelin Pekgün, A. Serdar Şimşek, and Şafak Yücel, as well as seminar participants at Bilkent University, Consortium Friday Virtual Seminar Series of Indiana University, Florida State University, North Carolina State University, University of Colorado Boulder, University of Rochester, and University of Waterloo for valuable comments. The authors are grateful to a few students from the University of Notre Dame and University of Texas at San Antonio, especially Rapha Roque, for helping clean and organize the data set.

Endnotes

¹ Until the end of the 2014 season, the NFL forbade televising in a team's local market if 85% of the seats were not sold three days prior to the start of the game.

² Except for the New Orleans Saints, all teams eventually adopted variable pricing.

³ For instance, an NFL game book for a 2014 regular season game can be obtained by clicking on the "Game Center" tab of that game on <http://www.nfl.com/schedules/2014/REG1> and downloading the corresponding pdf file.

⁴ See <https://archive.org/web/>.

⁵ See <https://teammktg.com> reports the weighted average of ticket prices for general seat categories, determined by factoring the ticket price for each category as a percentage of the total number of seats in each stadium.

⁶ See <https://projects.fivethirtyeight.com/complete-history-of-the-nfl/> provides the history of each team's Elo rating. See <https://fivethirtyeight.com/features/introducing-nfl-elo-ratings/> for the methodology.

⁷ For example, we obtained the movements before the 2014 season from <http://thesportsdaily.com/nfl-free-agency-2014-ranking-signings-moves-contracts-analysis-impact-2014-season/> through the Internet Archive's Wayback Machine search tool.

⁸ See <https://www.forbes.com/sites/forbespr/2014/08/20/forbes-announces-17th-annual-nfl-team-valuations/#264c11ed76c8>.

⁹ See <https://factfinder.census.gov/faces/nav/jsf/pages/index.xhtml>.

¹⁰ See <http://www.diversitydata.org/> provides the ethnic heterogeneity tables for each team's hometown. We use these tables to calculate the Gini coefficients for ethnic diversity.

¹¹ The NFL hosted nine NFL International Series games outside the United States in 2012–2015. The Buffalo Bills hosted one game at the Roger Center in Toronto in 2012 and 2013. Finally, the 2014 regular season game between the New York Jets and Buffalo Bills was played in Detroit due to a snowstorm in Buffalo.

¹² We thank the anonymous referee for suggesting this idea.

¹³ Due to renovations in their stadiums, some of the home games for the Miami Dolphins, the Cleveland Browns, the Oakland Raiders, and the Green Bay Packers are not used in the analysis.

¹⁴ As a robustness check, we performed the same analysis also in the quarter-season and home game levels and observed a similar pattern. Complete results are available from the authors upon request.

¹⁵ The proofs for all propositions are given in Online Appendix E.

¹⁶ To validate our results, we performed multiple robustness checks using different classification strategies. The details are given in Online Appendix F.

¹⁷ Because we do not have any detailed information on the seat category capacities, we are not able to calculate a weighted average. However, we observe that markdown and markup percentages are similar across different seat categories. Consequently, we believe that a weighted average would provide similar estimates.

¹⁸ Prior to estimation of these specifications presented in this section, we perform an identification check similar to the one in Section 4.3. The estimation results show that there is no statistically significant difference between the daily listing trends across the treatment and control groups. The details are given in Online Appendix G.

¹⁹ Please note that D_d becomes the index of the number of remaining weeks to the game.

²⁰ To reduce potential moderating effects of other demographics, we include additional demographic interactions, population, and ethnic diversity, with VP in Equation (8). Our findings remain robust with the inclusion of these interactions.

²¹ We manually count the price tiers for the teams that make their seating maps available on the Internet. We complement these data with the numbers reported in Courty and Davey (2020).

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